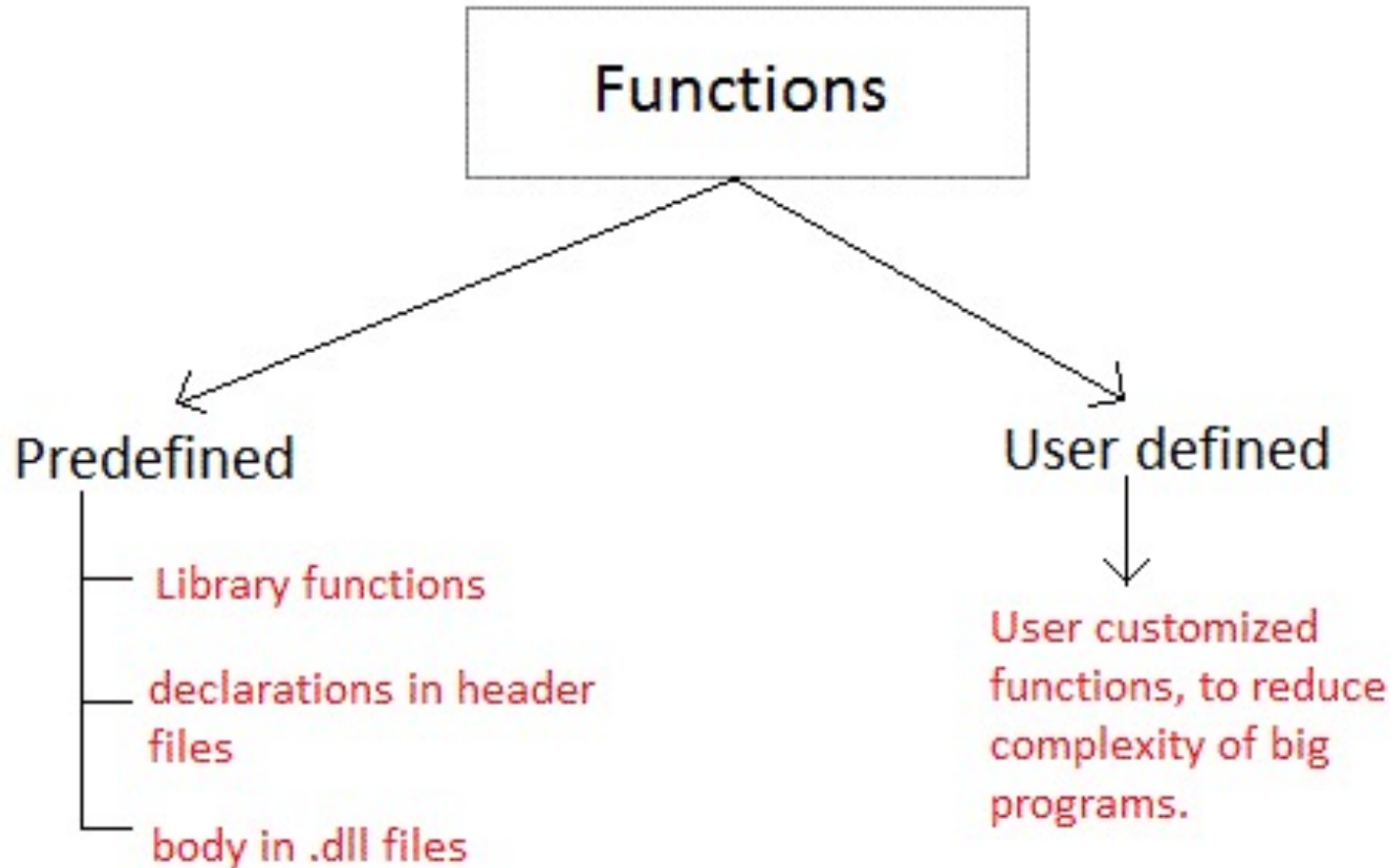


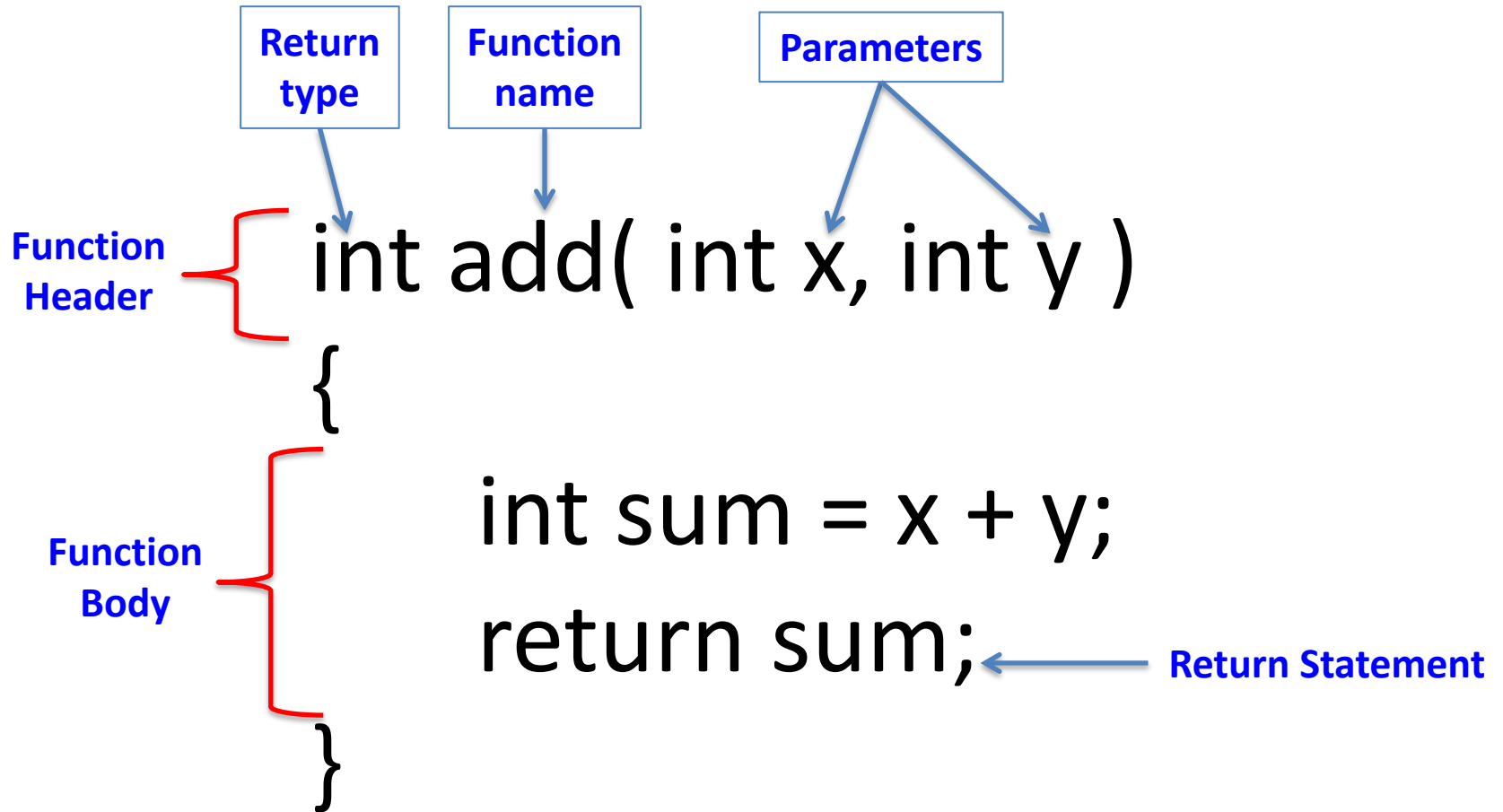
# **Library Functions and Header Files**

# Function

❖ **Function:** A function is a block of codes that performs specific tasks.



# Dissection of a Function



# How do connect Functions?

```
#include <stdio.h>
```

**Function prototype** → `int add(int x, int y);`

**Calling function**

```
int main()
{
    int n1, n2, result;
    .....
    result = add(n1, n2);
    .....
}
```

**Function call**

**Called function**

```
int add (int x, int y)
{
    int sum = x + y;
    return sum;
}
```

# Some predefined/library functions

- ❖ **Library:** A library in C programming is collection of functions and declarations.
- ❖ **Header file:** A header file contains function prototypes, identifier declarations and macros.

## Some important library functions

| Function    | Return Type | Purpose                          |
|-------------|-------------|----------------------------------|
| abs(i)      | int         | Return the absolute value of i   |
| fabs(d)     | double      | Return the absolute value of d   |
| sin(d)      | double      | Return the sine of d             |
| cos(d)      | double      | Return the cosine of d           |
| tan(d)      | double      | Return the tangent of d          |
| pow(d1, d2) | double      | Return d1 raised to the d2 power |

i = integer

d = double

# Some important library functions

| Function     | Return Type | Purpose                                |
|--------------|-------------|----------------------------------------|
| ceil(d)      | int         | Round up to the next integer value     |
| floor(d)     | int         | Round down to the next integer value   |
| round(d)     | int         | Round to the nearest integer value     |
| exp(d)       | double      | Raise e to the power d                 |
| log(d)       | double      | Return natural logarithm of d          |
| fmod(d1, d2) | double      | Return remainder of d1/d2              |
| sqrt(d)      | double      | Return the square root of d            |
| rand()       | int         | Return a random positive integer       |
| srand(u)     | void        | Initialize the random number generator |
| toascii(c)   | int         | Convert to ASCII                       |
| tolower(c)   | int         | Convert letter to lower case           |
| toupper(c)   | int         | Convert letter to upper case           |

i = integer

d = double

u = unsigned integer

c = character

# Some important library functions

| Function     | Return Type | Purpose                                 |
|--------------|-------------|-----------------------------------------|
| strlen(s)    | int         | Length of string without null character |
| strpos(s, c) | int         | Position of character c in sting s      |
| isdigit(c)   | 0/1         | Is c a digit?                           |
| isprint(c)   | 0/1         | Is c printable?                         |
| islower(c)   | 0/1         | Is c a lowercase letter?                |
| isupper(c)   | 0/1         | Is c a uppercase letter?                |
| isalnum(i)   | 0/1         | Is c an alphabet or numeric?            |
| isalpha(i)   | 0/1         | Is c an alphabet?                       |
| isascii(i)   | 0/1         | Is c an ASCII code?                     |

i = integer      d = double      u = unsigned integer      c = character      s = string

# Standard Input and Output (stdio.h)

## Some standard input functions

| Function  | Sample Example  | Purpose                                                                                    |
|-----------|-----------------|--------------------------------------------------------------------------------------------|
| getch()   | ch = getch();   | Read a character from keyboard and store it in variable "ch", does not show it on screen   |
| getchar() | ch = getchar(); | Read a character from keyboard and store it in variable "ch", show the character on screen |
| gets()    | gets(name)      | Read a string from keyboard and store it in memory location "name".                        |
| scanf()   | scanf("%d", &x) | Read integer from keyboard and store it in memory location x.                              |



# scanf() function

❖ **scanf()** function has two types of parameters: format and list of variables.

✧ `scanf( "%s %d %f", item, &partno, &cost);`  **No & is used for string**

✧ `scanf( "%d-%d-%d", &day, &month, &year);`  **Input shall be like 15-8-2020**

✧ `scanf( "%[ ABCDEFGHIJKLMNOPQRSTUVWXYZ]", line);`

**Input will be taken as long as uppercase letters are entered.**

✧ `scanf( "%[ ^\n]", line);`

**Input will be taken until newline character is entered.**

✧ `scanf( "%3d %3d %3d", &a, &b, &c);`

**Case 1:** If the input is 123 456 789, then a = 123 b= 456 c = 789.

**Case 2:** If the input is 123456789, then a = 123 b= 456 c = 789.

**Case 3:** If the input is 1234 5678 9, then a = 123 b= 4 c = 567.

✧ `scanf( "%3d %5f %c", &i, &x, &c);`

**If the input is 10 256.875 T, then i = 10 x= 256.8 c = T.**

# scanf() function Formatting Character

| Conversion Character | Meaning                                              |
|----------------------|------------------------------------------------------|
| c                    | Character                                            |
| d                    | Decimal integer                                      |
| hd or h              | Short integer                                        |
| ld                   | Long integer                                         |
| D                    | Long integer                                         |
| u                    | Unsigned integer                                     |
| o                    | Octal                                                |
| x                    | Hexadecimal                                          |
| i                    | Integer, decimal or hexadecimal or octal             |
| f                    | Floating point number                                |
| lf                   | Double                                               |
| e                    | Floating point in exponential or scientific notation |
| le                   | Double in exponential notation                       |
| s                    | string                                               |
| [...]                | String                                               |

# printf() function

❖ **printf()** function has two types of parameters: format and list of variables.

```
int i = 123456;
```

```
float x = 345.678;
```

✧ `printf("%3d %5d %9d", i, i, i)`



Output: **123456 123456 123456**

The entire integer will displayed if the field width is smaller

✧ `printf("%3f %10f %13f", x, x, x)`



**Six digits will be displayed after decimal point**

Output: **345.678000 345.678000 345.678000**

✧ `printf("%3e %13e %16e", x, x, x)`

Output: **3.456780e+02 3.456780e+02 3.456780e+02**

✧ `printf("%3g %13g %16g", x, x, x)`

Output: **345.678 345.678 345.678**

# printf() function

```
int a = 0x80ec;
```

```
float b = 0.3e-12, x = 345.678;
```

```
✧ printf("%4x %7d %10.2f %16e", a, a, x, x);
```

Output: 80ec 33004 345.68 3.456780e+02

```
✧ printf("%4X %10.2E %16.2e", a, b, x);
```

Output: 80EC 3.00E-13 3.46e+02

```
char line[12];
```

```
printf(":%15s %15.5s %.5s:\n", line, line, line);
```

```
printf(":%-15s %-15.5s %-.5s:\n", line, line, line);
```

**Output:**

: lower-case lower lower:

:lower-case lower lower:

# printf() function

```
int i = 1234, j=01777, k =0xa08c;
```

```
float x = 12.0, y = -3.3;
```

```
Printf(":%8u %8o %8x %7.0f %10.1e:\n", i, j, k, x, y);
```

```
Printf(":%-8u %-8o %-8x %-7.0f %-10.1e:\n", i, j, k, x, y);
```

```
Printf(":%+8u %+8o %+8x %+7.0f %+10.1e:\n", i, j, k, x, y);
```

```
Printf(":%-+8u %-+8o %-+8x %-+7.0f %-+10.1e:\n", i, j, k, x, y);
```

```
Printf(":%#8u %#8o %#8X:\n", i, j, k);
```

```
Printf(":%08u %08o %08X:\n", i, j, k);
```

```
Printf(":%7.0f %#7.0f %7g %#7g:", x, x, y, y);
```

## OUTPUT:

```
:      1234      1777      a08c      12      -3.3e+00:
: 1234      1777      a08c      12      -3.3e+00      :
:      +1234      +1777      +a08c      +12      -3.3e+00:
: +1234      +1777      +a08c      +12      -3.3e+00      :
:      1234      01777      0xA08C:
: 00001234 00001777 0000A08C:
:      12      12.      -3.3 -3.300000:
```

# printf() function

| Conversion Character | Meaning                                              |
|----------------------|------------------------------------------------------|
| c                    | Character                                            |
| d                    | Decimal integer                                      |
| hd or h              | Short integer                                        |
| ld                   | Long integer                                         |
| D                    | Long integer                                         |
| u                    | Unsigned integer                                     |
| o                    | Octal                                                |
| x                    | Hexadecimal                                          |
| i                    | Integer, decimal or hexadecimal or octal             |
| f                    | Floating point number                                |
| lf                   | Double                                               |
| e                    | Floating point in exponential or scientific notation |
| le                   | Double in exponential notation                       |
| s                    | string                                               |
| [...]                | String                                               |

# printf() function

| Flag | Meaning                                                                                                                                                                                                                              |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -    | Left justified                                                                                                                                                                                                                       |
| +    | A sign (either + or -) will precede each signed numeric data item                                                                                                                                                                    |
| 0    | Causes leading zeros                                                                                                                                                                                                                 |
| "    | Blank space                                                                                                                                                                                                                          |
| #    | <ul style="list-style-type: none"><li>✧ 0 and 0x will be displayed before octal and hexadecimal number</li><li>✧ A decimal point will be displayed in all floating point numbers, even if the data item is a whole number.</li></ul> |